

1. Let the n th triangular number $T_n = 1 + 2 + 3 + \cdots + n$ and

$$P_n = \frac{T_2}{T_2 - 1} \cdot \frac{T_3}{T_3 - 1} \cdots \frac{T_n}{T_n - 1}.$$

Find P_{2022} .

2. Compute the following sum:

$$2^2 \cdot 3! + 3^2 \cdot 4! + \cdots + 100^2 \cdot 101!$$

3. Let $a \star b$ be equal to one of $a + b$, $a - b$, $a \cdot b$, or a/b , each chosen with equal probability. If the expected value of

$$((((1 \star 2) \star 3) \cdots \star 9) \star 10)$$

can be written as $\frac{P_1 P_2 P_3 P_4}{P_5}$, for prime powers $P_1, P_2, \dots, P_5$, what is

$$P_1 + P_2 + \cdots + P_5?$$

4. Suppose the Fibonacci sequence begins with $F_1 = 1$, $F_2 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 3$. Find the minimal n such that

$$\sum_{i=2}^n \frac{1}{F_{i-1} F_{i+1}} \geq \frac{4095}{4096}.$$

5. Let the sequence $\{a_i\}_{i=0}^{\infty}$ be defined by $a_0 = \frac{1}{2}$ and $a_n = 1 + (a_{n-1} - 1)^2$. Find the product

$$\prod_{i=0}^{\infty} a_i = a_0 a_1 a_2 \dots$$

6. For a positive integer $k \geq 2$, let a_k , b_k and c_k be the complex roots of the equation

$$\left(x - \frac{1}{k-1}\right) \left(x - \frac{1}{k}\right) \left(x - \frac{1}{k+1}\right) = \frac{1}{k}.$$

Compute the sum

$$\sum_{k=2}^{\infty} \frac{(a_k^2 + a_k)(b_k^2 + b_k)(c_k^2 + c_k)}{k+1}.$$

7. Find the value of

$$\sum_{a=1}^{\infty} \sum_{b=1}^{\infty} \sum_{c=1}^{\infty} \frac{ab(3a+c)}{4^{a+b+c}(a+b)(b+c)(c+a)}.$$

8. Let $F_1 = 1$, $F_2 = 1$, and $F_{n+2} = F_{n+1} + F_n$ for $n \geq 1$. Find

$$\sum_{n=1}^{\infty} \arctan\left(\frac{1}{F_n}\right) \arctan\left(\frac{1}{F_{n+1}}\right)$$